Graphics Programming

Material Class

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Material Objects

- Store three types of data related to rendering
  - Shader program references
  - Uniform object
  - OpenGL render settings
- Will be many extensions of the base class
  - Different materials for rendering geometric data as a collection of points, as a set of lines, or a surface
  - Some basic materials will implement vertex colors of uniform base colors
  - Advanced materials will implement texture mapping, lighting, and other effects

Base Class - Tasks

- Compiling the shader code and initializing the program
- Initializing dictionaries to store uniforms and render settings
- Defining uniforms corresponding to model, view, and project matrices
- Defining a **locateUniforms** function to determine and store all the uniform variable references in the shaders
- Defining a **setProperties** function to set multiple uniform and render setting values simultaneously from a dictionary

Base Class - Extension

- Contain the actual shader code
- Add any extra uniform objects required by the shaders to the uniforms dictionary
- Call the **locateUniforms** function once all uniform objects have been added
- Add OpenGL render settings (as Python variables) to the setting dictionary
- Implement **updateRenderSettings** function that will call the OpenGL functions needed to configure the render settings

Basic Material Class

- Contains shader code and a set of corresponding uniforms
- Will be extended
  - PointBasicMaterial, LineBasicMaterial, SurfaceBasicMaterial
  - Each one will contain the relevant OpenGL render settings for the corresponding type of geometric primitive

Basic Material – Shader Variables

- Vertex Shader
  - Two attributes: vertex position and vertex color
  - Three Uniform variables: **Model, view, and projection matrices**
    - Used to calculate final position of each vertex
- Fragment Shader
  - We could interpolate vertex colors or apply a single color to all
  - Two Uniform variables
    - **vec3 baseColor** applied to all vertices
    - **Bool useVertexColor** apply vertex colors?

Point Basic Material

+Render settings

+"**drawStyle**"	GL_POINTS
+"**pointSize**"	integer
+"**roundedPoints**"	boolean

Line Basic Material

+Render settings

+"**drawStyle**"	GL_LINE_STRIP, GL_LINE_LOOP, GL_LINES
+"**lineType**"	"connected", "loop", "segments"
+"**lineWidth**"	integer

Surface Basic Material

+Render settings

+“**drawStyle**”	GL_TRIANGLES
+“**doubleSide**”	boolean
+“**wireframe**”	boolean
+“**lineWidth**”	integer